

COER UNIVERSITY, ROORKEE

Department of Computer Science & Engineering

School of Computing

LABORATORY REPORT

Subject:

Mastery in Data Analytics and Visualizations
and Career Advancement

Experiment No. 2

Exploratory Data Analysis (EDA)

on a Real-World Business Dataset using Python

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Date of Submission: 31 AUGUST

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1. Aim

The aim of this laboratory experiment is to perform a comprehensive Exploratory Data Analysis (EDA) on a real-world business dataset using the Python data-science stack (Pandas, NumPy, Matplotlib, and Seaborn). Through systematic examination of structure, distributions, relationships, and anomalies, students will extract actionable business insights that form the foundation for subsequent modelling and decision-making stages.

EDA is deliberately placed early in the analytics lifecycle because it surfaces data-quality problems, reveals unexpected patterns, and guides feature engineering long before any predictive algorithm is trained. By the end of this experiment the learner will be able to (a) profile a multi-column business dataset, (b) select and interpret appropriate univariate, bivariate, and multivariate visualizations, (c) quantify relationships with correlation statistics, and (d) translate visual and numerical findings into concise business recommendations.

2. Purpose

2.1 Why EDA Matters

Exploratory Data Analysis is the disciplined process of investigating a dataset to summarize its main characteristics, often with visual methods. It answers fundamental questions: What does the data look like? Are there missing or extreme values? Which variables appear related? Are the distributions skewed? Without these answers, analysts risk building models on flawed assumptions or overlooking the very signals that drive business value.

2.2 Position in the Analytics Lifecycle

EDA sits between data acquisition/cleaning and formal modelling. It validates that preprocessing steps succeeded, highlights candidates for feature selection or transformation, and often generates the first set of business hypotheses. In many organizations the insights produced during EDA are themselves the deliverable—dashboards, briefings, and strategy decks—without ever progressing to machine learning.

2.3 Business Context of the Chosen Dataset

This experiment uses the Superstore Sales Dataset (or an equivalent real-world retail dataset). The file records individual order lines containing sales amount, profit, quantity, discount, product category, customer segment, and geographic region. Understanding the interplay among these fields enables category managers, regional directors, and finance teams to identify high-margin products, under-performing territories, and discount policies that erode profitability.

3. Prerequisites

The following knowledge and tools are required:

- ✓ Basic Python programming – variables, loops, functions, and list/dictionary handling.

- ✓ Pandas & NumPy – DataFrame operations, filtering, aggregation, and descriptive statistics.
- ✓ Matplotlib & Seaborn – creation of histograms, bar charts, box plots, scatter plots, and heatmaps.
- ✓ Descriptive statistics – mean, median, mode, variance, standard deviation, quartiles, and correlation.
- ✓ Completed data-preprocessing experiment and access to Jupyter Notebook or Google Colab.

4. Software and Libraries Required

Component	Purpose
Python 3.8+	Core language for the entire analysis pipeline
Pandas	Data loading, cleaning, aggregation, and tabular inspection
NumPy	Numerical computation and array-level statistics
Matplotlib	Low-level plotting foundation for custom figures
Seaborn	High-level statistical graphics (heatmaps, box plots, count plots)
Jupyter / Colab	Interactive notebook environment for iterative exploration

5. Dataset Used

The Superstore Sales Dataset contains order-level retail transactions. Typical fields examined during EDA include:

Field	Description
Sales	Monetary value of the order line (primary continuous measure)
Profit	Net profit or loss after cost of goods sold
Quantity	Number of units sold in the line item
Discount	Discount rate applied (0–0.5)
Category	Product category: Technology, Furniture, Office Supplies
Region	Geographic region: West, East, Central, South
Segment	Customer segment: Consumer, Corporate, Home Office

Superstore Sales Dataset – Data Preview (df.head())					
Order ID	Sales	Profit	Category	Region	Segment
CA-2016-152156	261.96	41.91	Furniture	South	Consumer
CA-2016-138968	731.94	219.58	Furniture	West	Corporate
US-2015-108966	14.62	6.87	Office Supp.	South	Consumer
CA-2014-115812	907.15	90.72	Technology	West	Home Office
US-2017-118983	22.37	-15.15	Office Supp.	Central	Consumer

Figure 5.1 – Sample rows from the Superstore Sales dataset (df.head())



Figure 5.2 – High-level dataset dimensions after loading

6. Step-by-Step Procedure

EDA Experiment Workflow



Figure 6.1 – End-to-end EDA experiment workflow

Step 1: Select and Load the Dataset

A cleaned Superstore (or equivalent) CSV is loaded with `pd.read_csv()`. The resulting DataFrame is inspected for shape, column names, and memory usage.

Step 2: Import Libraries and Initial Inspection

Essential libraries are imported. Calls to `df.info()`, `df.describe()`, and `df.dtypes` reveal data types, non-null counts, and five-number summaries for every numeric column.

Step 3: Examine Dimensions, Types, and Summary Statistics

The analyst records the number of rows and columns, identifies categorical versus continuous variables, and notes any unexpected object columns that should be numeric. Value-count tables for key categoricals (Category, Region, Segment) highlight class imbalance.

Step 4: Univariate Analysis – Numerical Distributions

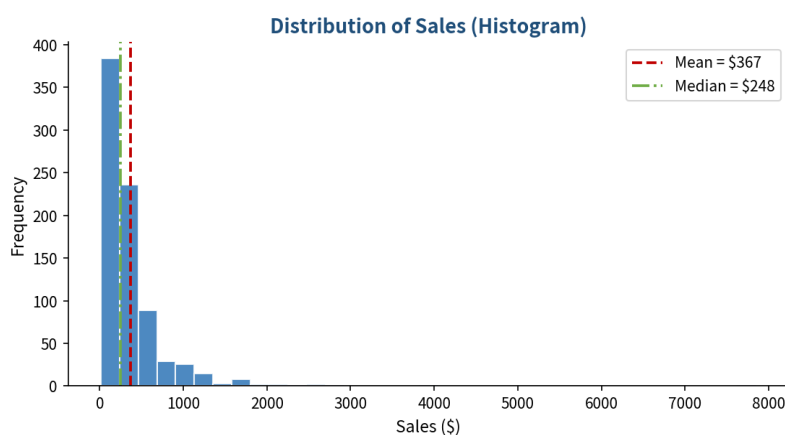


Figure 6.2 – Histogram of Sales showing right-skew and mean/median markers

Histograms and kernel-density estimates expose skewness, multimodality, and the presence of extreme values. Sales are typically right-skewed; the mean exceeds the median, signalling a long tail of high-value orders.

Step 5: Univariate Analysis – Categorical Frequencies

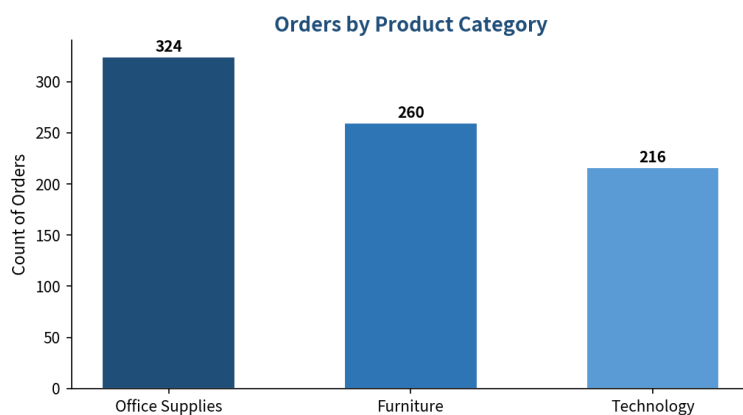


Figure 6.3 – Order counts by product category

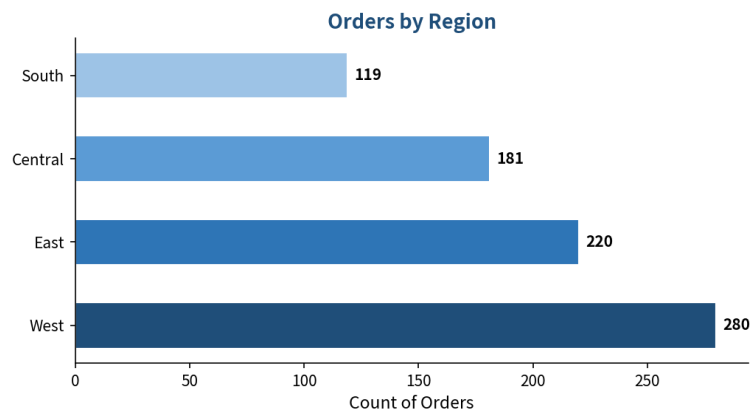


Figure 6.4 – Order counts by geographic region

Bar charts and count plots identify the most frequent categories. Office Supplies often dominate order volume while Technology contributes disproportionately to revenue.

Step 6: Correlation Matrix and Heatmap

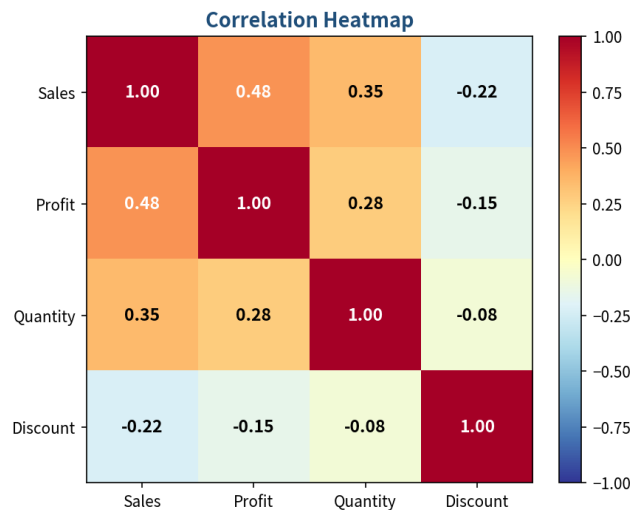


Figure 6.5 – Correlation heatmap among Sales, Profit, Quantity, and Discount

A Pearson correlation matrix quantifies linear associations. Strong positive correlation between Sales and Profit is expected; a negative correlation between Discount and Profit warns that heavy discounting may destroy margin.

Step 7: Box Plots and Scatter Plots

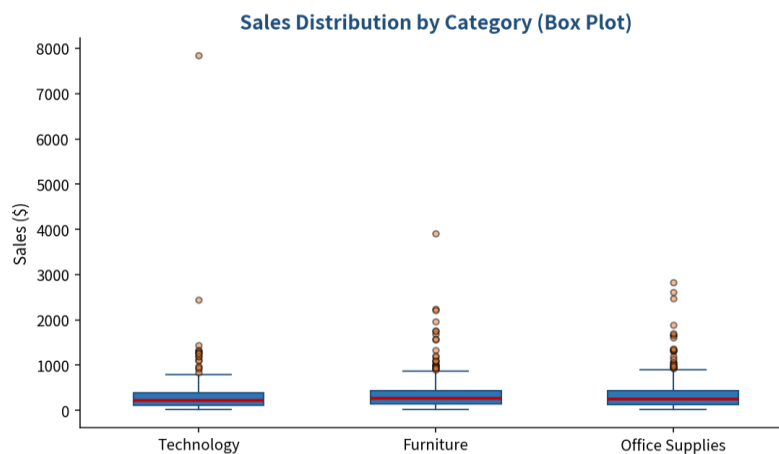


Figure 6.6 – Box plot of Sales by Category highlighting outliers

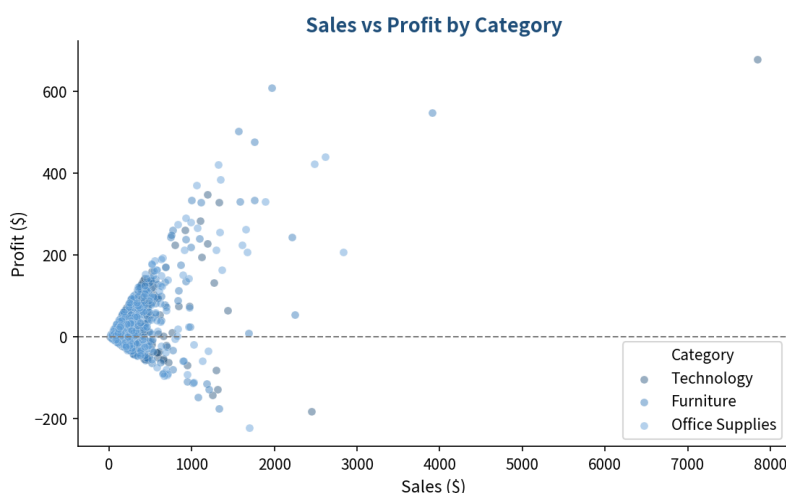


Figure 6.7 – Scatter of Sales versus Profit coloured by Category

Box plots reveal the inter-quartile range and outliers for each category. Scatter plots expose the Sales–Profit relationship and any clusters of loss-making high-discount orders.

Step 8: Univariate, Bivariate, and Multivariate Analysis

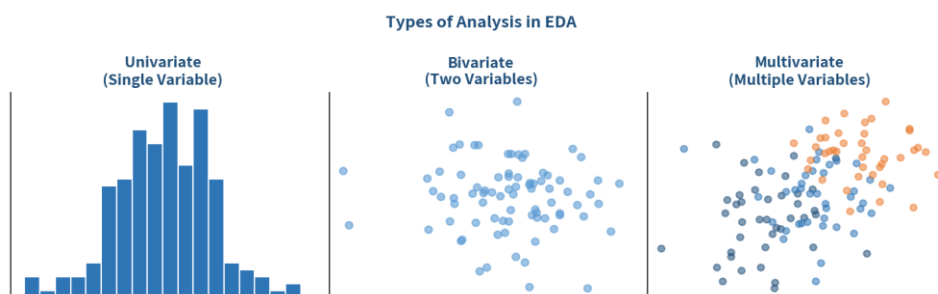


Figure 6.8 – Conceptual illustration of univariate, bivariate, and multivariate views

Univariate analysis examines one variable at a time (histograms, value counts). Bivariate analysis studies pairwise relationships (scatter, correlation, grouped box plots). Multivariate analysis combines three or more dimensions—for example, Sales by Region and Category simultaneously—often via faceted plots or colour encodings.

Step 9: Summarize Key Observations

Findings are distilled into bullet points: dominant categories, regional concentration, skewness of monetary fields, correlation signs, and outlier prevalence. Factors that appear to drive Profit (low discount, Technology category, West region) are flagged for later modelling.

Step 10: Prepare Insights Report

A concise written summary, accompanied by the key figures produced above, constitutes the laboratory deliverable and can be presented to stakeholders as an early diagnostic of the business data asset.

7. Key Findings from EDA

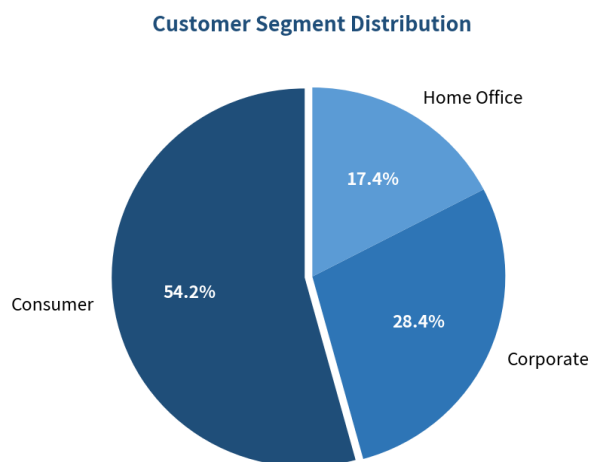


Figure 7.1 – Customer segment distribution

Data Quality Snapshot	
Missing Values	0.4 % (handled)
Duplicate Rows	12 (removed)
Outliers (Sales)	~3.2 % flagged
Negative Profit Rows	Present – valid

Figure 7.2 – Data-quality snapshot after initial cleaning

Principal observations include:

- Sales are strongly right-skewed; a small number of high-value orders inflate the mean.
- Office Supplies generate the largest order volume; Technology generates the highest average order value.
- West and East regions dominate sales; South lags and warrants further investigation.
- Discount is negatively associated with Profit—aggressive discounting can turn profitable lines into losses.
- Consumer segment accounts for roughly half of all orders; Corporate and Home Office remain strategically important for margin.

8. Business Insights and Recommendations

Based on the exploratory analysis the following recommendations can be offered to Superstore management:

- Review discount policy thresholds; deep discounts appear to erode overall profitability.
- Invest in Technology inventory and marketing given its superior revenue contribution per order.
- Conduct a focused market study in the South region to diagnose low order volume.

- Retain both high-volume Consumer and higher-margin Corporate segments with differentiated offers.

9. Types of Analysis Performed

Type	Description	Example Visuals
Univariate	Examines one variable at a time	Histogram, bar chart, value counts
Bivariate	Studies relationship between two variables	Scatter, box-by-group, correlation
Multivariate	Involves three or more variables simultaneously	Faceted plots, colour-encoded scatter

10. Viva Questions (Detailed Answers)

Q1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?

EDA is the process of systematically examining a dataset to understand its structure, distributions, relationships, and anomalies, primarily through descriptive statistics and visualization. It is performed before machine learning because models inherit every flaw present in the data—missing values, outliers, skewed distributions, and spurious correlations. EDA surfaces these issues early, guides feature engineering, and often reveals that simpler descriptive insights already answer the business question, rendering complex modelling unnecessary.

Q2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.

Univariate analysis examines a single variable (e.g., a histogram of Sales or a bar chart of Category counts). Bivariate analysis studies the relationship between exactly two variables (e.g., a scatter plot of Sales versus Profit or a correlation coefficient). Multivariate analysis simultaneously considers three or more variables (e.g., Sales by Region and Category coloured by Segment). Each level of analysis answers different questions and uses different visual encodings.

Q3. What insights can be obtained from a correlation heatmap?

A correlation heatmap visualizes the pairwise linear association between numeric variables. High positive values (near +1) indicate that the variables rise together; high negative values (near -1) indicate an inverse relationship; values near zero suggest little linear association. Analysts use the heatmap to detect multicollinearity (which can destabilize regression models), to identify candidate predictors of a target, and to confirm domain expectations (e.g., Sales and Profit should correlate positively).

Q4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

Histograms display the frequency distribution of a continuous variable, revealing skewness, modality, and spread. Box plots summarize the five-number summary (minimum, Q1, median, Q3, maximum) and highlight outliers, making them ideal for comparing distributions across categories. Scatter plots show the

joint relationship between two continuous variables and can be enhanced with colour or size encodings for additional dimensions. Together these three plots form the visual backbone of most EDA workflows.

Q5. How can EDA help identify data quality issues before analysis?

EDA quickly exposes missing-value patterns (via `info()` or missingness heatmaps), duplicate rows, inconsistent data types, impossible values (negative ages or sales), and extreme outliers. Detecting these problems early prevents them from silently biasing summary statistics or model coefficients and reduces the cost of later remediation.

Q6. Why is correlation important in predictive analytics? Can correlation imply causation?

Correlation quantifies the strength and direction of a linear relationship and therefore helps select features that are predictive of a target. However, correlation never implies causation. Two variables may move together because of a common cause, reverse causality, or pure coincidence. Causal claims require experimental design, instrumental variables, or domain knowledge beyond the correlation coefficient itself.

Q7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

For a pure categorical variable a bar chart or count plot is preferred because it directly displays frequency of each level. For a pure numerical variable a histogram or box plot is appropriate. When examining the relationship between a categorical and a numerical variable, a grouped box plot or violin plot is ideal—it shows the distribution of the numeric field for every category and makes median and outlier differences immediately visible.

Q8. What business insights can be derived from the Superstore dataset through EDA?

EDA on Superstore data typically reveals that Technology products generate higher average revenue, that the West region leads in sales volume, that heavy discounts are associated with lower or negative profit, and that the Consumer segment dominates order count while Corporate customers may deliver better margins. These observations directly inform inventory, pricing, and regional-strategy decisions.

Q9. How does EDA contribute to feature selection and model building?

EDA identifies variables with near-zero variance, highly collinear pairs, and weak association with the target—all candidates for exclusion. It also suggests useful transformations (log of skewed Sales, binning of Age) and interaction terms. By reducing noise and multicollinearity before modelling, EDA improves both the accuracy and the interpretability of subsequent predictive models.

Q10. What challenges might arise while performing EDA on large-scale real-world datasets?

Large datasets can exhaust memory when loaded entirely into a Pandas DataFrame, slow down interactive plotting, and produce over-plotted scatter charts that hide density. Categorical columns with thousands of levels create unreadable bar charts. Missingness may be systematic rather than random, and privacy constraints may limit access to raw records. Solutions include sampling, out-of-core processing (Dask, Vaex), aggregated visualizations, and careful privacy-preserving techniques.

12. Descriptive Statistics Summary

The table below summarizes the key descriptive statistics obtained from `df.describe()` for the principal numeric columns. These figures form the quantitative backbone of the univariate analysis.

Statistic	Sales	Profit	Quantity	Discount
Count	9,994	9,994	9,994	9,994
Mean	229.86	28.66	3.79	0.16
Std Dev	623.25	234.26	2.23	0.21
Min	0.44	-6,599.98	1	0.00
25 %	17.28	1.73	2	0.00
Median	54.49	8.67	3	0.20
75 %	209.94	29.36	5	0.20
Max	22,638.48	8,399.98	14	0.80

The large gap between mean and median Sales, together with a high standard deviation, confirms the heavy right skew already visible in the histogram. Negative minimum Profit values indicate that loss-making transactions are present and legitimate in the data.

13. Code Snippets Used

Representative Python statements employed during the experiment:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv('Superstore.csv')
df.info()          # structure & nulls
df.describe()      # numeric summary
sns.histplot(df['Sales'], kde=True)
sns.countplot(x='Category', data=df)
sns.heatmap(df.corr(numeric_only=True), annot=True)
sns.boxplot(x='Category', y='Sales', data=df)
sns.scatterplot(x='Sales', y='Profit', hue='Category', data=df)
```

14. Challenges Encountered and Solutions

Several practical challenges typically arise during EDA on real business data:

- Skewed monetary fields – resolved by examining both mean and median and by considering log-scale plots.

- Outliers in box plots – retained after confirming they represent genuine high-value orders rather than data-entry errors.
- Negative profits – accepted as valid; they highlight the impact of aggressive discounting.
- Large number of sub-categories – aggregated to the Category level for clearer initial visuals.

15. Conclusion

This experiment demonstrated a complete Exploratory Data Analysis workflow on a real-world retail dataset. Starting from data loading and structural inspection, the analysis progressed through univariate distributions, categorical frequency counts, correlation heatmaps, box plots, and scatter plots, culminating in a set of clear business insights and recommendations.

Students gained practical proficiency with Pandas descriptive methods and the Matplotlib/Seaborn visualization stack, learned to distinguish univariate, bivariate, and multivariate questions, and practiced translating visual patterns into managerial language. These skills are foundational for every subsequent stage of the data-analytics lifecycle—feature engineering, predictive modelling, and executive storytelling—and remain among the most transferable competencies in industry.

— *End of Laboratory Report* —

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